TEXORA · LECTURE SHEET

Knitting Elements and Motions

Every knitting calculation in the sheets that follow counts things this sheet defines: needles, sinkers, cams and the loop-forming cycle they perform together. Master the elements and their motions first — courses, feeders and production rates are just this cycle, counted.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 7 · SPECIALIZATION — FABRIC MANUFACTURING

1. The elements of a weft knitting machine

ELEMENT	FUNCTION
Needle	Forms the loop. Three types: **beard (spring-beard)** — hook closed by a presser, fine gauges, warp knitting and older frames; **latch** — hook closed by its own pivoted latch, the universal choice for weft knitting; **compound** — hook closed by a sliding closing element, for high-speed fine work.
Sinker	Thin blade between needles: holds the fabric down (holding-down), supports the yarn at knock-over, and shapes the sinker loop. On circular machines the sinker ring rotates with the cylinder.
Cams	Fixed tracks that move the needle butts: **stitch (knock-over) cam** sets how far the needle descends — and therefore the stitch length; **clearing (raising) cam** lifts needles to clear old loops; **guard cams** keep butts in track. Cam setting is stitch length setting.
Feeder	Presents yarn to the needles at one knitting station; courses per revolution = feeders on the machine.
Take-down & winding	Draws the formed fabric away at a controlled rate (fabric tension affects stitch length) and winds or folds it.
Needle bed / cylinder	Carries the needles in tricks (slots); circular machines have a cylinder (vertical needles) and often a dial (horizontal needles) at right angles to it.

2. The loop-forming cycle — motions in order

One stitch is made by a choreographed sequence, identical in principle on a hand frame and a 90-feeder circular machine:

1. **Clearing (raising):** the cam lifts the needle; the old loop slides down the stem and opens the latch, coming to rest below it.
2. **Feeding (yarn presentation):** the feeder lays the new yarn across the open hook while the sinkers hold the fabric.
3. **Latch closing:** as the needle starts to descend, the old loop — held by the sinker — slides up the stem and flips the latch shut over the new yarn.
4. **Knock-over (casting off):** the needle descends through the old loop; the old loop slips over the closed hook and is cast off, while the new yarn is drawn through it.
5. **Loop formation complete:** the descending needle pulls the new loop to its final size against the knock-over edge — the depth of descent at this instant *is* the stitch length, which is why the stitch cam is the machine's master setting.

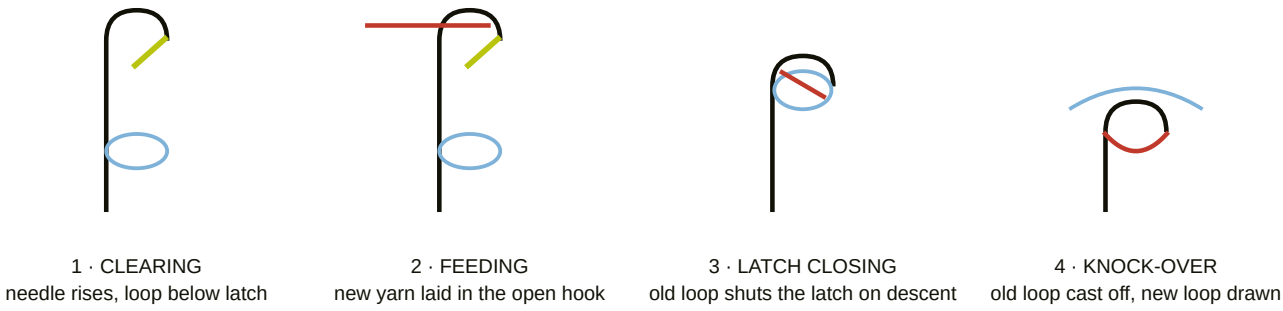


Figure 1 — The latch-needle cycle in four positions: clearing, feeding, latch closing, knock-over. Needle: black; latch: green; old loop: blue; new yarn: red.

3. Gauge and needle count — the first calculation

Gauge is needles per inch of bed. On a circular machine the total needle count follows from diameter and gauge:

$$\text{Total needles} = \pi \times \text{diameter (in)} \times \text{gauge}$$

WORKED EXAMPLE — NEEDLE COUNTS

30-inch, 24-gauge cylinder: $N = \pi \times 30 \times 24 =$ **2,262 needles** (the machine of Sheet 8).

26-inch, 28-gauge: $N = \pi \times 26 \times 28 =$ **2,287 needles** (the machine of Sheet 12).

One course on the 24-gauge machine at stitch length 3.0 mm consumes $N \times l = 2{,}262 \times 3.0$ mm = **6.79 m of yarn** — the seed number from which all circular-knitting production arithmetic grows.

4. Motions beyond the single needle

Patterning motions interrupt the cycle selectively: a needle held down **misses** (float), a needle stopped part-way makes a **tuck** (yarn held uncast with the old loop), and full descent **knits** — the three loop types of TE-3101/3205, produced here by cam selection (pattern wheels, drums, or electronic actuators) acting butt by butt. **Machine motions** complete the picture: cylinder rotation carries needles past fixed cam boxes (on most circular machines the needles move, the cams stand still; the fabric rotates and is taken down through the cylinder's centre), feeders stand fixed around the circumference, and each feeder–cam-box pair is one **system** knitting one course per revolution — the reason Sheet 8 multiplies rpm by feeders to get courses per minute.

Glossary

Gauge — needles per inch of needle bed. **Clearing** — raising the needle so the old loop passes below the latch. **Knock-over** — casting the old loop off over the closed hook. **Stitch cam** — the cam whose setting fixes needle descent and hence stitch length. **System/feeder** — one knitting station; one course per machine revolution per system. **Trick** — the slot in the bed that guides one needle.

Key points

• Elements: needle, sinker, cams, feeder, take-down — each with one job in the cycle.
• Cycle order is fixed: clearing → feeding → latch closing → knock-over; stitch length is set at knock-over by the stitch cam.
• Needles = π × diameter × gauge (2,262 and 2,287 in the course's two reference machines); yarn per course = needles × stitch length.
• Miss and tuck are the same cycle, interrupted at different heights by selection.

Practice

1. Find the needle count of a 20-inch, 18-gauge cylinder.
2. On that machine at stitch length 4.2 mm, how much yarn does one course consume?
3. A fabric shows long floats on the face where loops should be. Which motion has failed, and at which step of the cycle?
4. Why does doubling the feeders of a circular machine double its courses per minute but not the length of yarn in one course?

Answer key

1. N = π × 20 × 18 = **1,131 needles**.
2. 1,131 × 4.2 mm = 4,750 mm ≈ **4.75 m per course**.

3. The needles are **missing**: they were not raised to clearing height (selection/cam fault at Step 1), so no new loop was drawn and the yarn floated across them.
4. Yarn per course is set by the cylinder alone — needles × stitch length — and feeders do not change either. Feeders add *stations*: each revolution now knits one course per feeder, so courses per minute (and yarn per minute) multiply while the single-course length is untouched.
